

Summary

ESSENTIAL QUESTION

How can we investigate and apply our understanding of energy transfer and transformation?



TOPIC 3.1:

What are the properties of energy?

- Energy can produce change in a system.
- There are different forms of energy.
- Energy can be transferred or transformed.
- Physical quantities contribute to different forms of energy.

Key Terms

system
surroundings
kinetic energy
potential energy
law of conservation of energy



TOPIC 3.2:

How is energy transformed?

- Energy is transformed in chemical reactions.
- Energy is transformed in nuclear reactions.
- Energy is transformed when light energy interacts with matter.

Key Terms

radiation	nuclear decay
nuclear fission	chain reaction
nuclear fusion	absorption



TOPIC 3.3:

How does energy transformation affect global systems?

- Earth is a system in which energy is transformed.
- Earth's atmosphere is heated by the transformation and transfer of solar energy and thermal energy.
- Energy transfer and energy transformation moderate Earth's temperature.
- Energy transformation and transfer can harm aquatic and terrestrial ecosystems.

Key Terms

conduction

convection

specific heat capacity



TOPIC 3.4:

How does energy transformation affect humans?

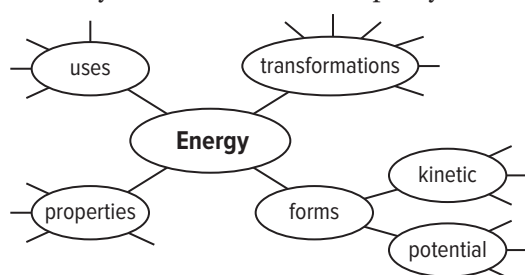
- Energy transformations provide benefits, but not without risk of harm.
- We take action and create technology to reduce the harmful impacts of energy transformation.
- We learn from our mistakes and each other to reduce harmful impacts of energy transformation.

Review

What Do You Know? Connecting to Concepts

Visualizing Ideas

- Expand the following map about energy. Add as many extensions to the map as you can.



- Create a Venn diagram that compares kinetic energy with potential energy.

Using Key Terms

- Use the list of key terms below to create three or four categories into which the terms can be divided. Write the names of the categories and list the key terms that fit into each. Explain why the terms fit into each category.

- chain reaction
- conduction
- convection
- kinetic energy
- law of conservation of energy
- nuclear decay
- nuclear fission
- nuclear fusion
- potential energy
- radiation
- specific heat capacity
- surroundings
- system

Communicating Concepts

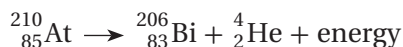
- List three forms of potential energy that we transform into electrical energy.
- Describe two ways you can indirectly observe energy.
- The photo shows a scene at the Capilano Suspension Bridge Park in Vancouver. Identify as many forms of energy as you can that are present in or that could be inferred from the photo. Identify each form as kinetic or potential.



- In living things, glucose reacts with oxygen to produce energy, according to the reaction below.

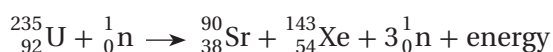
$$\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{energy}$$
 If energy is released in this reaction, how can photosynthetic organisms carry out the reverse of this reaction and combine carbon dioxide and water to make glucose?
- ATP is often involved when energy transformations occur in living things.
 - What is ATP?
 - Identify two processes that involve ATP.
 - For each example you listed in part b), explain the role of ATP.

9. Use the following nuclear equation to answer the questions below.



- a) What type of nuclear reaction is occurring in the equation?
 - b) What is represented by ${}_2^4\text{He}$?
10. a) Explain how the mass number of an atom can remain the same if its atomic number changes in a nuclear reaction.
- b) In what type of reaction(s) does this happen?

11. Use the following nuclear equation to answer the questions below.



- a) What type of reaction does this equation represent?
 - b) Where would you expect to find this type of reaction occurring?
 - c) Describe one characteristic of the products of this reaction.
12. Compare chemical reactions and nuclear reactions in terms of how much energy they transform.
13. a) Identify two different atmospheric greenhouse gases.
- b) Describe one positive effect of these gases.
- c) Describe one negative effect of the gases.
14. a) What is methane cracking?
- b) What is steam methane reforming?
- c) Which of the above processes is more harmful to the environment? Explain why this is the case.
15. Over history, we have developed technologies that have helped us benefit from energy transformations, but these technologies have also had negative effects. Use two examples to support this statement.

What Do You Know?

Connecting to Competencies

Developing Skills

16. a) Write the equation that describes gravitational potential energy.
- b) Use this equation to create a graph that shows how gravitational potential energy of an object changes as height measured from a reference point increases.
17. Use a sketch or diagram to explain how energy is transformed and transferred to bring a pot of water to a boil on a stove.
18. **Figure 3.32** shows how cells in the retina of the eye transform light energy to enable vision. Use an analogy to help explain this process to a grade 7 student.
19. Use a diagram or write a description to show the relationships among evaporation, gravitational potential energy, transpiration, solar energy, precipitation, thermal energy, and condensation in terms of energy transformation.

Thinking Critically and Creatively

20. It has been said, "The only truly isolated system is the universe itself." What does that statement mean? Explain whether you believe that it is accurate, and why.
21. In a battery, chemical reactions transform chemical potential energy into electrical potential energy. When attached to a circuit, the electrical potential energy is transformed into electrical kinetic energy in an electric current. A load can transform this electrical kinetic energy to other forms of energy. Explain how fuel cells are similar to batteries, and how they are different.

Unit 3 Review *(continued)*

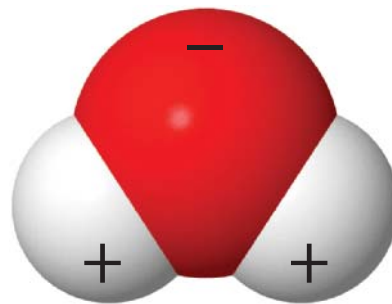
- 22. a)** Use an analogy to explain the concept of specific heat capacity to a student who is not taking this course.
- b)** Describe an observation you might make that would tell you that the specific heat capacity of water is higher than the specific heat capacity of land. Justify your response.
- 23.** You give your friend a push on a swing. A short time later, your friend and the swing come to a halt.
- a)** Identify a system and its surroundings in this example.
- b)** Explain why the law of conservation of energy can still be applied to this example. Refer to the system and surroundings you identified in part **a)** in your response.
- 24.** Photovoltaic cells are made of two thin slices of silicon treated with different materials. They are then sandwiched together. Why could a single slice of silicon not be used instead?
- 25.** Hydrogen, helium, and traces of lithium were produced as part of the early history of the universe, before stars formed. Atoms of all other elements were produced by fusion reactions that happened in stars or in supernova explosions. Scientists of First Peoples, Western traditions, and other ways of knowing commonly describe humanity as being made of “star stuff.” What does this description mean?
- 26.** Uncontrolled fission reactions can result in a reactor meltdown at a nuclear power plant. Control rods in a nuclear reactor control the rate at which fission reactions occur. They are lifted from and lowered into the reactor as needed. The control rods contain cadmium, which absorbs neutrons. How do you think control rods control the rate at which fission reactions occur?

- 27.** Photovoltaic cells and vision are two means through which energy is transformed when light interacts with matter. Describe another example in which matter absorbs light and energy is transformed.
- 28.** All solar energy that reaches Earth’s surface is re-radiated back into space, mostly as infrared radiation. However, solar energy maintains life on Earth. Explain how both these things are possible.

Understanding Big Ideas Making New Connections

Applying Your Understanding

- 29.** Water has unique properties that make it important to living organisms. One of these properties is a large specific heat capacity relative to many other substances. Study the image of a water molecule. Notice that one end is negatively charged and the other end is positively charged. Think about how water molecules interact with each other. Now, use the concepts you are thinking about to suggest why water has such a large specific heat capacity.



- 30.** Why do scientists find it so difficult to develop a way to control nuclear fusion so it can be used to generate electrical energy?

Thinking Critically and Creatively

- 31.** Most take-out pizza boxes are made of cardboard.
- a)** Use the terms open system, closed system, and isolated system to explain why pizza boxes do not keep a pizza very warm.
 - b)** Use your knowledge of these different systems to suggest a design for an improved pizza box.
- 32.** Identify three things you did today that used energy. For each example, describe how energy was transferred or transformed.
- 33.** A scientist has developed a building material that reduces the damage caused by earthquakes. Use the concepts of energy transfer and/or transformation to suggest how this material might work.
- 34.** Active solar-collector systems are often used to provide thermal energy for water heaters, pools, and homes. A simple active solar collector consists of a solar collector, a pump, a heat storage system, and a system of pipes to move the heat from one place to another. Solar energy heats water that circulates through pipes. The hot water is pumped to the location that is to be heated, such as a pool. The cold water is pumped back through the pipes so it passes through the solar collectors, once again being heated by energy from the Sun.
- a)** Describe how energy is transferred and transformed by an active solar collector.
 - b)** Active solar collectors have few negative effects on the environment. Do you think this technology would be a good way to heat your school? Explain your reasoning.

Connecting Self and Society

- 35.** Only as much as 10% of the electrical kinetic energy that is used by a halogen light bulb is transformed into light energy.
- a)** What happened to the remaining 90% of the energy?
 - b)** LED bulbs are more efficient than halogen bulbs, but they are expensive to buy. Would you choose to buy LED bulbs? Explain why or why not.
 - c)** Do you think most of society would answer part **b)** in the same way you did? Explain.
- 36.** Could society survive without energy transformations? Explain your reasoning.
- 37. a)** When an accident happens at a nuclear power plant, the local environment can take a long time to recover from the damage. Explain why.
- b)** When a fossil-fuel-burning power plant is operating normally, it contributes pollutants to the atmosphere. Explain how this can affect living things.
 - c)** Compare the above negative effects as well as any others you know of related to the generation of electrical energy from nuclear reactions and fossil fuels. Which one do you think has the greatest negative effect on society overall? Explain your reasoning.
- 38.** Review the five themes that are the foundation of First Peoples perspectives of science. How do these themes relate to Western science perspectives on energy transformation, energy transfer, and the law of conservation of energy? Write, record, or narrate your ideas, along with reasons in support of them.



Unit Assessment

How do sustainable vehicles compare to conventional vehicles?



A current push toward the development and use of sustainable vehicles around the world is helping to reduce emissions of greenhouse gases and other pollutants. These vehicles typically do not rely on the combustion of fossil fuels. Instead, scientists and engineers have turned to other energy transfers and transformations—for instance, the transfer of the mechanical kinetic energy of the wind, and the transformation of solar energy and chemical potential energy of non-fossil fuel sources. Some sustainable vehicles do use fossil fuels, but in a way that reduces the impact of their combustion. Sustainable vehicles are not problem-free, however. Issues to resolve include factors such as a speed, cost, availability, public acceptance, and potential effects on living things and the environment.

Work as part of a group to do the following.

- STEP 1** ► Reflect on the three options, their photos, and the questions asked in their captions.
- STEP 2** ► Brainstorm at least three more options and questions of your own about sustainable versus conventional vehicles.
- STEP 3** ► Decide on one of the six questions to investigate.
- STEP 4** ► Plan and conduct a scientific inquiry to explore your question.
- STEP 5** ► Organize and analyze the data and information that you find and collect.
- STEP 6** ► Communicate the results of your inquiry in a suitable manner.

OPTION A

Solar in the Sky

How can a model help you demonstrate that solar aircraft are, or are not, a practical alternative to conventional aircraft?





OPTION B

Electric Vehicles

UBC is actively engaged in using, as well as further developing, electric cars. How do electric cars compare to gas-powered ones?

OPTION C

Fuel-cell Safety

How do fuel-cell vehicles compare to conventional ones in terms of safety for their human passengers?



Assessment Criteria

Did I and my group...

- Develop one or more questions that provided opportunities for rich investigation? **QP**
- Develop effective methods to collect and record reliable data and information? **PC**
- Analyze, reflect on, and draw meaningful conclusions as related to the inquiry? **PA**
- Evaluate the process and results of the inquiry, troubleshooting problems if they arose? **E**
- Compare at least one type of sustainable vehicle with a conventional vehicle? **AI**
- Present the results of the inquiry using language, conventions, and representations appropriate for a specific purpose and audience? **C**